

Environmental Dependence of Galaxy Evolution: A comparison of Cluster and Field Galaxies

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ABSTRACT

I examine three observational methods: optical spectroscopy, optical and near-infrared imaging, and HI 21 cm radio observations, to assess evidence that galaxy evolution proceeds differently in clusters compared to the field. Focusing on optical spectroscopy, I review three independent studies. Results from [Rines et al. \(2005\)](#) and [Balogh et al. \(1997\)](#) show reduced star-formation rates in cluster environments, driven by suppressed H α and [O II] emission in cluster cores, with activity increasing toward larger radii. In contrast, [Bretherton et al. \(2010\)](#) find that tidally disturbed cluster spirals can exhibit enhanced H α emission, in some cases exceeding field levels. Together, these studies indicate that although cluster galaxies do indeed evolve differently from field galaxies, the environmental influence is not governed by a simple radial trend but instead reflects a combination of environmentally driven quenching and interaction-triggered enhancement.

USE OF GENERATIVE AI TOOLS

Portions of this work were assisted by the use of OpenAI’s ChatGPT. GenAI support was used for the following tasks: improving grammatical errors and structure, generating LaTeX-compatible formatting in accordance with AAS_{Te}X structure, and assistance with creating a bibliography file for citations. All scientific interpretation, methodological choices, data analysis, and final conclusions were made by me.

1. INTRODUCTION

Galaxy evolution refers to the changes in a galaxy’s stellar content, gas reservoirs, morphology, and star-formation activity over cosmic time. Understanding galaxy evolution is important because it provides insight into the physical processes shaping the universe, helping us to predict galaxy populations, interpret observational data, and refine cosmological models. Modern studies of galaxy evolution rely

heavily on large, multiwavelength surveys to track population changes across cosmic time. As described by the Max-Planck Institute, astronomers use ground and space-based telescopes such as the Very Large Telescope (VLT), Hubble Space Telescope (HST), and Herschel to observe galaxies over a wide redshift range ([Genzel 2013](#)).

The environment plays a significant role in shaping galaxy evolution. Galaxy clusters contain high galaxy number densities, deep gravitational potentials, and a hot intracluster medium (ICM), which differ substantially from the low-density field. Observations show that cluster galaxies are generally redder, less actively star-forming, and more morphologically evolved than similar-mass field galaxies, indicating that environmental processes can alter a galaxy’s gas content and star-formation history ([Kuchner et al. 2017](#)).

This paper aims to examine three observational approaches that provide evidence that galaxy evolution proceeds differently in clusters: (1) optical spectroscopy, which traces star-formation rates and gas properties; (2) optical and near-infrared imaging, which reveals structural and stellar population differences; and (3) radio observations of HI 21 cm, representing cold gas reservoirs that accelerate star formation from which we can infer the effect of quenching and environmental stripping on cluster galaxies. Optical spectroscopy will be the primary focus, with a detailed review of selected studies used to compare star-formation activity and stellar populations in cluster and field galaxies.

2. METHODS

2.1. Optical Spectroscopy

2.1.1. Description

Optical spectroscopy measures the light from galaxies dispersed into its component wavelengths and is a fundamental tool for studying galaxy evolution, particularly for assessing how environment affects star-formation activity. By measuring emission and absorption lines, astronomers can determine star-formation rates (SFRs), identify recently quenched populations, and estimate stellar population ages. Comparing these properties between cluster and field galaxies allows direct investigation of environmental influences on galaxy evolution.

2.1.2. Data Acquisition

Spectroscopic data are obtained using multi-object spectrographs on large ground-based telescopes, which enable simultaneous observations of multiple galaxies. [Balogh et al. \(1997\)](#) used optical spectroscopy to measure [O II] $\lambda 3727$ lines of 15 X-ray luminous clusters. [Rines et al. \(2005\)](#) measured H α emission in local clusters ($z < 0.1$), covering galaxies from dense cores out to the surrounding infall

regions. [Bretherton et al. \(2010\)](#) obtained integrated H α and [N II] fluxes for low-redshift cluster galaxies and a matched field sample. These surveys provide spectra with sufficient resolution to accurately measure line fluxes and equivalent widths, allowing for effective comparisons between cluster and field environments.

2.1.3. Analysis

Key measurements derived from optical spectroscopy include emission lines such as H α and [O II] $\lambda 3727$, which trace ionizing photons from young massive stars and provide quantitative estimates of ongoing star-formation rates. Cluster galaxies often show weaker or absent emission lines compared to field galaxies of similar mass and redshift, indicating suppressed or quenched star formation. Absorption features, particularly Balmer lines like H δ , identify recently quenched stellar populations ([Balogh et al. 1997](#)). Environmental effects are determined by comparing cluster galaxies to field galaxies. For example, [Balogh et al. \(1997\)](#) found a lower fraction of emission-line galaxies in clusters, while [Rines et al. \(2005\)](#) observed radial gradients in H α emission, showing that the dense cluster environment directly reduces star-formation activity.

2.1.4. Assumptions and Uncertainty

Spectroscopic analyses assume that emission lines reliably trace star-formation rates, with minimal contamination from non-stellar processes. Balmer absorption features are assumed to primarily reflect recent quenching rather than variations in metallicity or older stellar populations.

In cluster galaxies, the absence or weakness of emission lines is interpreted as quenched star formation, but some galaxies may have low-level or obscured star formation that is not detected. Dust extinction can interfere with observed line fluxes, potentially leading to underestimated star-formation rates. If not accounted for prop-

erly, there is also a possibility of contamination of emission lines from background galaxies, resulting in biased values for star formation.

2.2. *Optical and Near-Infrared (NIR) Imaging*

2.2.1. *Description*

Optical and near-infrared (NIR) imaging captures galaxies across multiple wavelengths to study their morphology, structure, and stellar populations. By comparing imaging data from cluster and field galaxies, researchers can quantify differences in size, shape, color, and luminosity, which reflect variations in star-formation history and quenching. Imaging studies provide spatially resolved data that complements spectroscopy, allowing both structural and population analyses. This method was used in Dressler’s groundbreaking research pertaining to the morphology-density relation for galaxies (Dressler 1980).

2.2.2. *Data Acquisition*

Imaging data are obtained from ground and space-based telescopes. Dressler (1980) used optical imaging from the Palomar 5m Hale telescope to observe 55 rich clusters, enabling classification of galaxies as elliptical, S0, or spiral. Nantais et al. (2013) used Hubble Space Telescope (HST) imaging of the cluster RX J0152.7–1357 at $z = 0.84$ in multiple optical bands, producing high-resolution images suitable for detailed light-profile fitting and morphological analysis. Data are reduced and calibrated to provide flux- and color-calibrated images for quantitative comparison between cluster and field galaxies.

2.2.3. *Analysis*

Morphologies are classified either by visual inspection (Dressler 1980) or by automated light-profile fitting (Nantais et al. 2013), which provide structural parameters such as Sérsic index, bulge-to-disk ratio, and effective radius. Colors derived from multi-band imaging trace stellar

ages and metallicities, with redder galaxies indicating older, more quenched populations. Comparisons between cluster and field galaxies reveal clear environmental trends. Dressler (1980) found that the fraction of elliptical and S0 galaxies increases with local density, while spirals dominate lower-density field environments. Nantais et al. (2013) observed a higher fraction of early-type, bulge-dominated galaxies in clusters compared to the field at similar redshifts, confirming that dense environments influence galaxy structure and stellar populations.

2.2.4. *Assumptions and Uncertainty*

Imaging analyses assume that galaxy morphology reflects evolutionary state and that colors primarily trace stellar age and metallicity. Classification can be subjective, particularly for intermediate or distant galaxies. Structural parameters depend on image resolution and signal-to-noise, and low-mass or high-redshift galaxies may be poorly resolved. Dust extinction can redden galaxies and complicate interpretations of stellar populations.

2.3. *H I 21 cm Observations*

2.3.1. *Description*

H I 21 cm observations are radio wavelength observations used to measure the neutral atomic hydrogen content of galaxies, which represents the cold gas reservoirs that fuel future star formation. This method provides a direct way to study how environmental effects in clusters influence the availability of gas. By comparing the H I content of galaxies in clusters and in the field, we obtain direct evidence of environmental gas depletion, which suppresses star formation in clusters.

2.3.2. *Data Acquisition*

H I 21 cm data are obtained using radio telescopes capable of detecting the 21 cm emission line from neutral hydrogen. Observations may be conducted with single-dish instruments, such

as Arecibo or the Green Bank Telescope, which measure the total H I flux from a galaxy, or with radio interferometers, such as the Very Large Array (VLA) or MeerKAT, which provide spatially resolved maps of the H I distribution.

2.3.3. *Analysis*

H I 21 cm line fluxes are converted into neutral hydrogen masses, representing the cold gas reservoirs of galaxies. We can analyze the H I line profile to study gas kinematics, including rotation and asymmetries. Cluster galaxies often show H I deficiency, truncated disks, or asymmetric gas distributions, which indicate environmental effects such as ram-pressure stripping or tidal interactions. By comparing H I masses and line profiles between cluster and field galaxies of similar stellar mass, researchers can quantify the impact of the environment on gas content and the potential for future star formation, directly linking cluster membership to suppressed stellar formation rates (Cortese et al. 2011).

2.3.4. *Assumptions and Uncertainties*

H I 21 cm observations assume that neutral hydrogen traces the galaxy’s cold gas reservoir, though molecular hydrogen (H_2) is not detected and may contribute significantly to the total gas content. Sensitivity limits can prevent the detection of low-H I galaxies, potentially underestimating environmental effects. Spatial and spectral resolution can obscure small-scale gas asymmetries or detailed kinematics, and the method is generally limited to low-redshift galaxies ($z < 0.2$) due to the faintness of the 21 cm signal. However, it is still reasonably accurate at these low redshifts.

3. RESULTS

In this section I will be focusing on optical spectroscopy as my method, and reviewing three studies (Rines et al. 2005; Balogh et al. 1997; Bretherton et al. 2010) that investigate

star formation activity in cluster and field galaxies. All three methods primarily focus on $\text{H}\alpha$ and $[\text{O II}]$ emission, but apply them to different redshift ranges and cluster environments. Together, these works provide complementary evidence on how the environment influences galaxy evolution.

3.1. *CAIRNS: Environmental Dependence of H-alpha Properties of Galaxies (Rines et al. 2005)*

3.1.1. *Overview*

Rines et al. (2005) investigate how star formation activity varies with environment by analyzing $\text{H}\alpha$ emission in galaxies across the virial region (region surrounding the cluster core), infall region (where galaxies are just beginning to enter the cluster), and the field using optical spectroscopy. Their sample spans several low-redshift clusters from the CAIRNS survey, providing high-completeness spectroscopy over large radial extents. By comparing the equivalent-width distributions and the relative fractions of emission-line galaxies, Rines et al. demonstrate that cluster environments strongly suppress star formation inside the virial radius with a gradual increase towards the infall, before finally reaching field-like levels beyond the virial radius. The study highlights the role of strong cluster cores on quenching star formation activity.

3.1.2. *Observational Results*

Trends in fraction of bright galaxies with and without $\text{H}\alpha$ emission as a function of radius

From figure 1, we see that the star forming fraction increases steadily with radius, where cluster cores have the lowest fraction of $\text{H}\alpha$ -emitting galaxies, while at $R/R_{200} \sim 1$, the infall region, the star-forming fraction converges to the field value. This trend can be observed across 8 different clusters as shown in the figure.

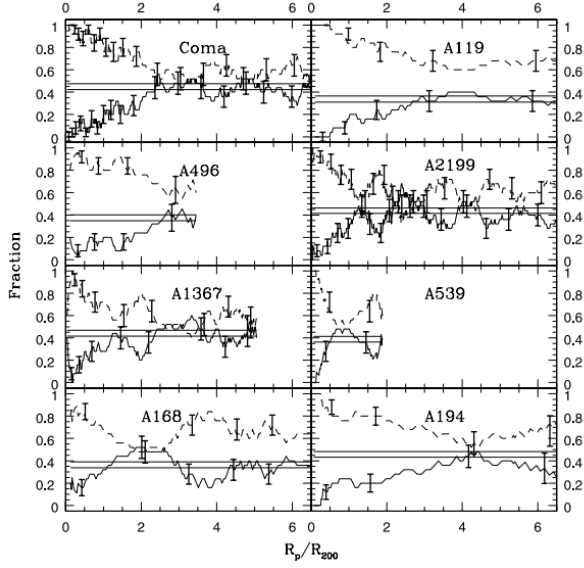


Figure 1. Fraction of bright galaxies with (solid lines) and without (dashed lines) $H\alpha$ emission as a function of projected clustercentric radius in units of R_{200} , with error bars indicating Poisson uncertainties in independent bins. Horizontal lines show the 1σ range of emission-line fractions in field samples matched to the same luminosity limit. Figure retrieved from [Rines et al. \(2005\)](#).

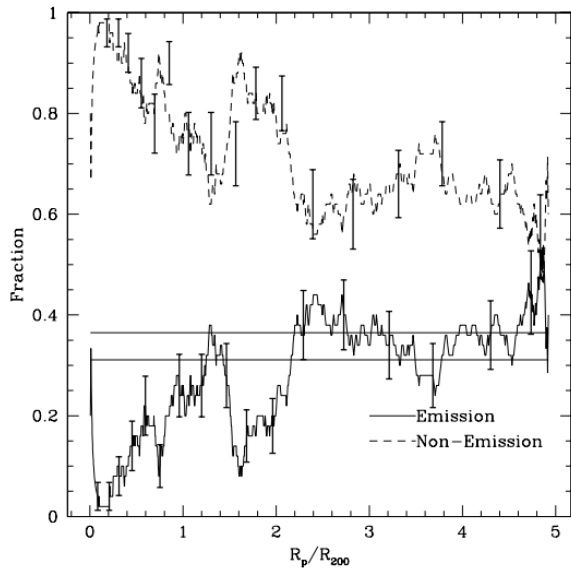


Figure 2. Fraction of bright galaxies from Figure 1, composited for all eight clusters. Figure retrieved from [Rines et al. \(2005\)](#).

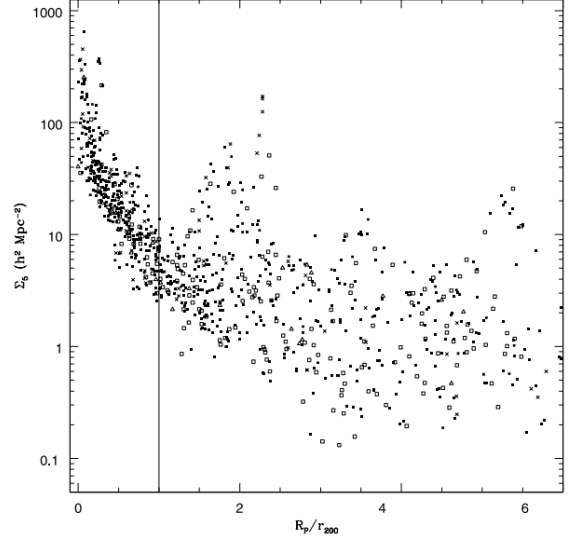


Figure 3. Local galaxy density versus projected clustercentric radius for bright galaxies in all CAIRNS clusters. Symbols show different spectral types: filled squares (absorption-dominated), open squares (star-forming), open triangles (AGN), and crosses (no FAST spectra). The vertical line marks R_{200} , separating the virial and infall regions. Figure retrieved from [Rines et al. \(2005\)](#).

Figure 2 composites the results from figure 1 to clearly show the trend between the fraction of $H\alpha$ emitting galaxies with radius persists across a number of clusters. The figure also includes poisson bars in order to show that the suppression is significant and not due to sampling variance.

Local Density Dependence

From figure 3, we observe a clear trend in absorption-dominated galaxies within the infall region, with an increase of star forming galaxies as we go beyond. We also see that beyond the infall region, the fraction of star forming galaxies is higher in lower density regions, indicating that star forming galaxies occupy lower density, larger radius regions.

$H\alpha$ Equivalent Width Distributions

From figure 4, we can see a clear divergence between cluster and field galaxies between 2 and 4 $EW(H\alpha)$, with infall and field galaxies exhibiting higher distributions for increasing $EW(H\alpha)$.

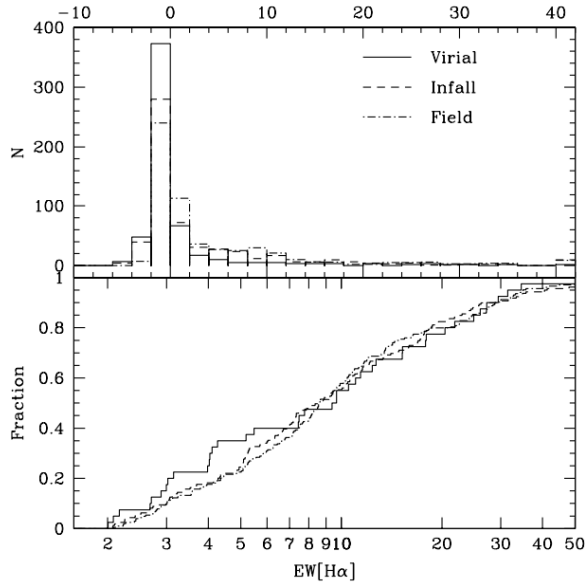


Figure 4. Distribution of H α equivalent widths for galaxies in the virial region, infall region, and the field. The top panel shows normalized histograms for each environment, while the bottom panel shows the cumulative EW(H α) distributions for galaxies with EW(H α) > 2 Å. Figure retrieved from Rines et al. (2005).

From this we see that cluster environments suppress both intensity and frequency of star formation.

3.1.3. Uncertainties and Assumptions

Interpreting the environmental trends presented in Rines et al. (2005) requires acknowledging several sources of uncertainty and the key simplifying assumptions built into their analysis.

The study assumes uniform absolute-magnitude selection across clusters, with each system cut at a slightly different M_{K_s} limit to ensure comparable completeness. Although this minimizes luminosity bias, it still introduces small differences in the sampled stellar-mass ranges between clusters, which can influence the star-forming fraction.

Projection effects introduce one of the dominant uncertainties. Because the survey measures projected clustercentric distances on the

sky, some galaxies located physically outside the virial radius can still appear projected within it. This “infall interloper” contamination systematically raises the apparent fraction of star-forming galaxies in the inner regions and complicates the interpretation of radial trends. The authors partially address this with kinematic modeling, but residual contamination remains.

Finally, uncertainties arise from cluster-centric scaling and profile fitting. The use of R_{200} to normalize distances assumes spherical symmetry and virial equilibrium, which may not perfectly describe real clusters.

However, keeping these uncertainties in mind, we can still see the correlation between galaxy evolution and environment.

3.2. Star Formation in Cluster Galaxies at $0.2 < z < 0.55$ (Balogh et al. 1997)

3.2.1. Overview

Balogh et al. (1997) analyzed a large spectroscopic sample of ~ 2000 galaxies drawn from 15 CNOC1 clusters spanning redshifts $0.18 < z < 0.55$, along with a matched sample of field galaxies. Their study investigates whether cluster galaxies show reduced star formation compared to field galaxies at intermediate redshift, using [O II] $\lambda 3727$ equivalent width as the primary measurement, from which star formation rates were derived. By directly comparing radial trends, and contrasting them with galaxies in field environments from the same sample, they were able to quantify the role of the environment in star formation rates of galaxies.

3.2.2. Observational Results

Cumulative [O II] Equivalent Width and Star Formation Rate(SFR) Distributions

The top panel of figure 5 shows that galaxies located in the cluster core (large dashed lines) have the lowest [O II] emission strengths, with their cumulative distribution shifted far left relative to both the outer cluster and field samples.

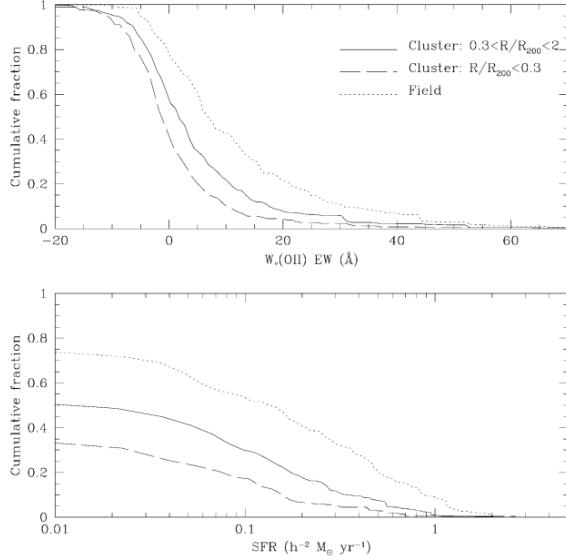


Figure 5. Top: Cumulative distribution of [O II] equivalent widths for inner cluster, outer cluster, and field galaxies. Bottom: Cumulative distribution of star formation rates for the same three populations, derived from [O II] using the Kennicutt (1992) relation. Figure retrieved from Balogh et al. (1997).

This indicates a significant suppression of ongoing star formation in the central cluster environment, which is shown clearly in the bottom panel, which plots the cumulative distribution of Star formation rates. These plots indicate the relative difference between cluster and field evolution

Radial Trend in Strong [O II] Emission

Referring to figure 6, we see the fraction of strongly emitting galaxies is lowest near the cluster center, rising steadily with increasing radius. By $R/R_{200} = 2$, the fraction begins to approach the field value, shown as the short-dashed horizontal comparison line.

The figure also includes a “color-predicted” fraction based on the known color–radius relation, shown as the long-dashed line. The observed fraction of strong [O II] emitters falls well below the color-predicted curve at virtually all radii, showing that suppression of star formation cannot be explained by galaxy color gradi-

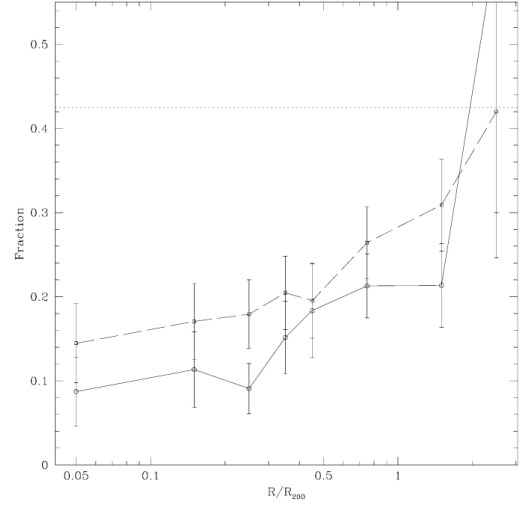


Figure 6. Fraction of cluster galaxies with $W_0([O II]) > 10$ Å as a function of projected radius, compared with the fraction expected from the color–radius relation. The field fraction (short dashed line) is shown for reference. Figure retrieved from Balogh et al. (1997).

ents alone, indicating that environmental processes directly reduce [O II] emissions, beyond what would be predicted from stellar populations alone.

3.2.3. Uncertainties and Assumptions

For the paper, the equivalent width of [O II], which was used as the primary tracer for star formation activity, was derived using the Kennicutt calibration, which was derived for nearby galaxies, and so a universal relation between [O II] strength and $H\alpha$ luminosity was assumed.

Uncertainties could occur in line strength measurements since [O II] is sensitive to dust extinction, and ionization. The study assumes [O II] is a reliable and unbiased indicator, however makes note of these biases.

There also uncertainties that could occur during projection of galaxy positions onto the clustercentric radius, which cannot distinguish in-fall galaxies from those projected onto it. This is minimized by using spectroscopic redshifts for

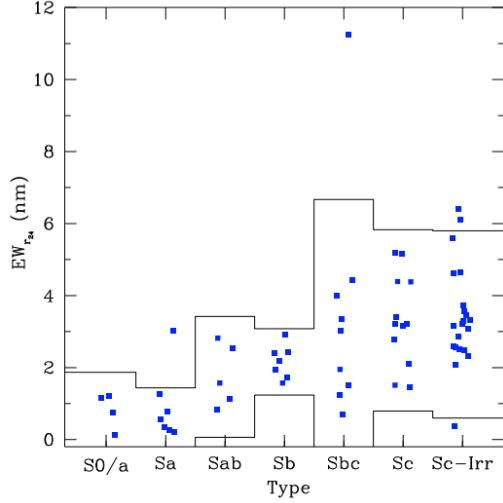


Figure 7. Global H α equivalent width as a function of morphological type for the field comparison sample. The solid line shows the 2σ limits of the field population. Figure retrieved from Bretherton et al. (2010).

selecting galaxies, however this uncertainty cannot be avoided for radial comparisons.

3.3. *Star-Forming Galaxies in Low-Redshift Clusters (Bretherton et al. 2010)*

3.3.1. Overview

Bretherton et al. (2010) examine spiral galaxies (Sa–Sc types) in six low-redshift clusters, comparing their global H α equivalent widths to field spirals. A key result is that tidal disturbances, rather than clustercentric radius, dominate the variation in H α emission. Their analysis emphasizes the role of galaxy–galaxy interactions and cluster substructure in influencing star formation activity.

3.3.2. Observational Results

Field vs. Cluster H α EW comparison (Figure 7 and Figure 8)

Figure 7 shows the distribution of H α equivalent widths (EW) for field spiral galaxies as a function of morphological type. This provides a baseline reference for what constitutes “nor-

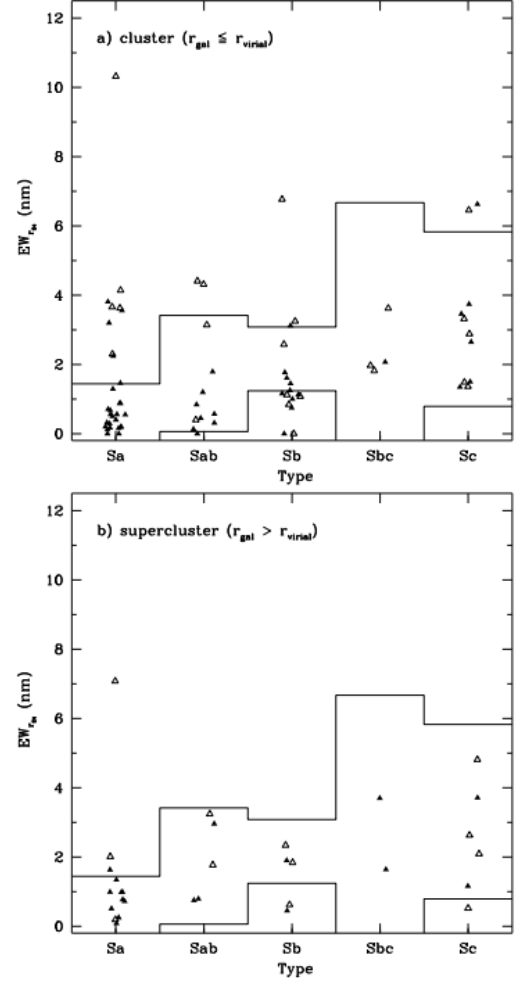


Figure 8. Equivalent width of H α as a function of Hubble type for cluster spiral galaxies, separated into disturbed (open triangles) and undisturbed (filled triangles) systems. Solid lines show the 2σ field limits from Figure 7. Figure retrieved from Bretherton et al. (2010).

mal” star formation in unperturbed spirals of types Sa, Sb, and Sc.

Figure 8 presents the corresponding EW(H α) measurements for cluster spirals. From these, we see a significant number galaxies, mainly in the Sa-Sb types exceeding the 2σ field limit from Figure 7, implying enhanced star formation rates. Most of these galaxies were also found to be classified as disturbed, indicating a correlation between star formation rate, and tidal disturbances.

3.3.3. *Uncertainties and Assumptions*

Classification of disturbed versus undisturbed spirals introduces a subjective element, as the identification of tidal features depends on the imaging quality and the criteria used. In addition, dust attenuation can reduce measured $H\alpha$ EWs, potentially lowering the apparent star formation rate. Cluster-to-cluster variations in environment, dynamical state, and galaxy population may also introduce scatter in the results. Despite these uncertainties, the separation between disturbed and undisturbed spirals remains clear and statistically significant.

4. CONCLUSIONS

The combined evidence from optical spectroscopy across three independent studies demonstrates that galaxy evolution proceeds differently in cluster environments compared to the field, but the manner of this evolution varies with the specific physical processes affecting the galaxies.

Rines et al. (2005) demonstrate a clear environmental trend: the fraction of $H\alpha$ -emitting galaxies increases steadily with radius, with cluster cores hosting the lowest star-forming fractions and the infall region converging to field values. Their $H\alpha$ EW distributions confirm that both the frequency and intensity of star formation are reduced in virialized regions, highlighting the role of cluster-scale mechanisms such as gas stripping and quenching.

Balogh et al. (1997) show that even when controlling for morphology using B/T classes, clus-

ter galaxies exhibit systematically lower $[O II]$ EW and lower star-formation rates than structurally similar field galaxies. The study also shows that outer-cluster galaxies retain higher star-formation activity, suggesting that quenching begins during the infall region, and intensifies as we move towards the core.

Bretherton et al. (2010) gives us a more contrasting, albeit complementary viewpoint: while many undisturbed spirals show reduced $H\alpha$ equivalent widths compared to the field, cluster spirals with strong tidal disturbances exhibit enhanced star formation, in some cases surpassing field levels. This indicates that tidal interactions and harassment can locally elevate star-formation rates, even within an environment that generally suppresses gas reservoirs, indicating that it is not a simple relation between clustercentric radius and star formation, but also dependent on other factors.

The most significant uncertainties arise from projection effects, sample completeness limitations, and emission-line selection biases, all of which may affect the inferred strengths of environmental trends.

However, noting all of these, these studies clearly show that cluster galaxies undergo a complex environmentally altered stellar evolution, where some galaxies experience cluster-driven quenching that suppresses star formation, while others show tidally driven enhanced star formation, both of which differ significantly from stellar evolution of field galaxies.

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